

CH 2 — Limits

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Limits

2.1 Definition of Limits

Definition — Neighborhood

An r -**neighborhood** of a point $(a, b) \in \mathbb{R}^2$ is a set

$$N_r(a, b) = \{(x, y) \in \mathbb{R}^2 : \|(x, y) - (a, b)\| < r\}$$

Note:

$\|(x, y) - (a, b)\|$ is the **Euclidean distance** or $\sqrt{(x - a)^2 + (y - b)^2}$

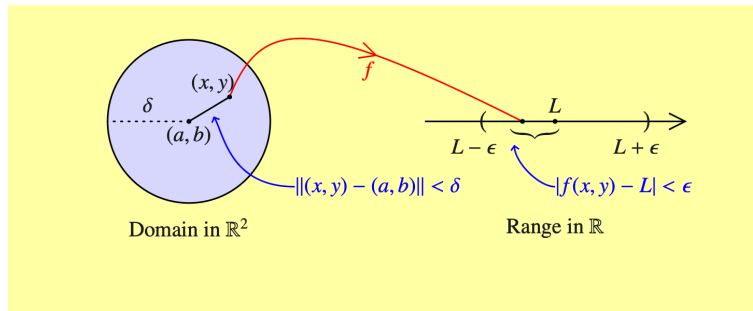
Visualize that neighborhood as an area of (x, y) that is the interval for a 3D plane

Definition — Limits

Assume that $f(x, y)$ is defined in some neighborhood of (a, b) , except possibly at (a, b) . Then if for every $\varepsilon > 0$, $\exists \delta > 0$ s.t.

$$0 < \|(x, y) - (a, b)\| < \delta \longrightarrow |f(x, y) - L| < \varepsilon$$

We say that $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$



2.2 Limit Theorems

Limit Theorems

If $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ and $\lim_{(x, y) \rightarrow (a, b)} g(x, y)$ both exist, then:

- $\lim_{(x, y) \rightarrow (a, b)} [f(x, y) \pm g(x, y)] = \lim_{(x, y) \rightarrow (a, b)} f(x, y) \pm \lim_{(x, y) \rightarrow (a, b)} g(x, y)$
- $\lim_{(x, y) \rightarrow (a, b)} [f(x, y)g(x, y)] = \left[\lim_{(x, y) \rightarrow (a, b)} f(x, y) \right] \left[\lim_{(x, y) \rightarrow (a, b)} g(x, y) \right]$
- $\lim_{(x, y) \rightarrow (a, b)} \frac{f(x, y)}{g(x, y)} = \frac{\lim_{(x, y) \rightarrow (a, b)} f(x, y)}{\lim_{(x, y) \rightarrow (a, b)} g(x, y)}$, for any $\lim_{(x, y) \rightarrow (a, b)} g(x, y) \neq 0$
- If $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ exists, then it is also unique

Proof

Uniqueness of a limit

Assume that $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L_1$ and $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L_2$

$$L_1 - L_2 = \lim_{(x,y) \rightarrow (a,b)} f(x,y) - \lim_{(x,y) \rightarrow (a,b)} f(x,y) = \lim_{(x,y) \rightarrow (a,b)} [f(x,y) - f(x,y)]$$

$$\implies \lim_{(x,y) \rightarrow (a,b)} 0 = 0 \implies L_1 - L_2 = 0 \implies L_1 = L_2$$

□

2.3 Proving a Limit denoted

In $\mathbb{R}^2$ we can approach the point (a,b) along an infinite number of paths.

For the limit to exist, f must approach L along **ALL** paths

Examples:

1. Let $f(x,y) = \frac{y^2 - x^2}{y^2 + x^2}$, prove $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \text{DNE}$

Try approaching $(0,0)$ along straight lines of the form $y = mx$

Along the line $y = mx$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{x \rightarrow 0} f(x, mx) = \lim_{x \rightarrow 0} \frac{m^2 x^2 - x^2}{m^2 x^2 + x^2} = \lim_{x \rightarrow 0} \frac{m^2 - 1}{m^2 + 1} = \frac{m^2 - 1}{m^2 + 1}$$

f approaches different values depending on m , so the limit does not exist as it is not unique.

2. Let $f(x,y) = \frac{|x|}{|x| + y^2}$; $(x,y) \neq (0,0)$, show that $\lim_{(x,y) \rightarrow (0,0)} f(x, mx) = 1 \ \forall m \in \mathbb{R}$, but $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \text{DNE}$

Try approaching $(0,0)$ along straight lines of the form $y = mx$

Along the line $y = mx$,

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{x \rightarrow 0} f(x, mx) = \lim_{x \rightarrow 0} \frac{|x|}{|x| + m^2 x^2}$$

$$\implies \lim_{x \rightarrow 0^-} \frac{-x}{-x + m^2 x^2} = \lim_{x \rightarrow 0^-} \frac{-1}{-1 + m^2 x} = 1$$

$$\implies \lim_{x \rightarrow 0^+} \frac{x}{x + m^2 x^2} = \lim_{x \rightarrow 0^+} \frac{1}{1 + m^2 x} = 1$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$$

Try approaching $(0,0)$ along $x = 0$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{x \rightarrow 0} f(0,y) = 0 \neq 1$$

Note: the above approach cannot be used to prove the existence of limit.

3. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ does not exist. (**Exponent matching**)

Try approaching $(0,0)$ along the straight lines of the form $y = mx$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 (mx)}{x^4 + (mx)^2} = \lim_{(x,y) \rightarrow (0,0)} \frac{mx}{x^2 + m^2} = 0, \ \forall (x,y) \neq (0,0)$$

Try approaching $(0,0)$ along $y = x^2$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 (x^2)}{x^4 + x^4} = \lim_{(x,y) \rightarrow (0,0)} \frac{1}{2} = \frac{1}{2}, \ \forall (x,y) \in \mathbb{R}^2$$

Notice that $0 \neq \frac{1}{2}$. Thus the limit does not exist.

Definition – Squeeze Theorem

If there exists a function $B(x, y)$ s.t.

$$|f(x, y) - L| \leq B(x, y), \quad \forall (x, y) \neq (a, b)$$

in some neighborhood of (a, b) and

$$\lim_{(x, y) \rightarrow (a, b)} B(x, y) = 0$$

then

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$$

Proof

Let $\varepsilon > 0$.

Since $\lim_{(x, y) \rightarrow (a, b)} B(x, y) = 0$, $\exists \delta > 0$ s.t. $0 \leq \|(x, y) - (a, b)\| \leq \delta \rightarrow |B(x, y)| < \varepsilon$

So, whenever $0 \leq \|(x, y) - (a, b)\| < \varepsilon$, if $|f(x, y) - L| < B(x, y) = |B(x, y)| < \varepsilon$,

and so by definition $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$

□

Example;

Show that $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 y}{x^2 + y^2} = 0$

We have $f(x, y) = \frac{x^2 y}{x^2 + y^2}$ and $L = 0$

$$|f(x, y) - L| = \left| \frac{x^2 y}{x^2 + y^2} \right| = \frac{x^2}{x^2 + y^2} |y| \leq \frac{x^2 + y^2}{x^2 + y^2} |y| = |y|, \quad \forall (x, y) \neq (0, 0)$$

$$B(x, y) = |y|$$

Notice $\lim_{(x, y) \rightarrow (0, 0)} B(x, y) = \lim_{y \rightarrow 0} |y| = 0$.

By Squeeze Theorem, $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$

Note: add to the numerator and do not subtrat from the denominator because it will change the domain of the functon.

Tip – Useful Inequalities for Upper Bound

1. $|a + b| \leq |a| + |b|, \quad a, b \in \mathbb{R}$

2. If $c \geq 0 \implies a \leq a + c$

Common use case, typically combined with Squeeze Theorem: $|x| = \sqrt{x^2} \leq \sqrt{x^2 + y^2}$

3. $2|x||y| \leq x^2 + y^2$

Note: The Squeeze Theorem requires to have a limiting value L in mind.

Example:

Determine $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - |x| - |y|}{|x| + |y|}$

Try $y = mx$

$$\lim_{x \rightarrow 0} \frac{x^2 - |x| - |mx|}{|x| + |mx|} = \lim_{x \rightarrow 0} \frac{|x| - 1 - |m|}{1 + |m|} = -1$$

We now want to prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - |x| - |y|}{|x| + |y|} = -1$

$$\left| \frac{x^2 - |x| - |y|}{|x| + |y|} + 1 \right| = \left| \frac{x^2 - |x| - |y| + |x| + |y|}{|x| + |y|} \right| = \frac{x^2}{|x| + |y|} \leq \frac{|x|(|x| + |y|)}{|x| + |y|} = |x|, \quad \forall (x, y) \neq (0, 0)$$

$$\lim_{(x,y) \rightarrow (0,0)} B(x, y) = \lim_{x \rightarrow 0} |x| = 0$$

By Squeeze Theorem, $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - |x| - |y|}{|x| + |y|} = -1$